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QoS Prediction for Satellite-Based Avionic Communication Using Transformers

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ABSTRACT Satellite-based communication systems are crucial for providing high-speed data services in aviation, particularly for business aviation operations that demand global connectivity. These systems face challenges from numerous interdependent factors, such as satellite handovers, congestion, flight maneuvers, and seasonal variations, making accurate Quality of Service (QoS) prediction complex. Currently, there is no established methodology for predicting QoS in avionic communication systems. This paper addresses this gap by proposing machine learning-based approaches for pre-flight QoS prediction. Specifically, we leverage transformer models to predict QoS along a given flight path using real-world data. The model takes as input a variety of positional and network-related features, such as aircraft location, satellite information, historical QoS, and handover probabilities, and outputs a predicted performance score for each position along the flight. This approach allows for proactive decision-making, enabling flight crews to select the most optimal flight paths before departure, improving overall operational efficiency in business aviation. Our proposed encoder-decoder transformer model achieved an overall prediction accuracy of 65% and an RMSE of 1.91, representing a significant improvement over traditional baseline methods. While these metrics are notable, our model's key contribution is a substantial improvement in prediction accuracy for underrepresented classes, which were a major limitation of prior approaches. Additionally, the model significantly reduces inference time, achieving predictions in 40 seconds compared to 6,353 seconds for a traditional KNN model. This approach allows for proactive decision-making, enabling flight crews to select optimal flight paths before departure, improving overall operational efficiency in business aviation.

INDEX TERMS Avionic communications, quality of service (QoS), machine learning, transformers, predictive modeling.

I. INTRODUCTION

THE increasing reliance on high-speed data services in aviation has underscored the pressing need for reliable and efficient communication systems that cater to both operational requirements and passenger demands on flights [1]. In the business aviation sector, this need is even more pronounced due to its niche challenges, such as irregular traffic patterns, diverse flight routes, and higher variability in user demands compared to commercial airlines. Satellite-based communication networks remain a cornerstone of connectivity for business jets, offering global coverage and high data throughput, even in remote regions and over transoceanic routes [2].

While there has been significant advancement in low Earth orbit (LEO) constellations, the business aviation sector continues to rely heavily on Geostationary (GEO) satellites. A key reason for this reliance is the reduced frequency of handovers in GEO satellite networks, which enables the use of mechanically steered antennas mounted on the aircraft tail. In contrast, LEO constellations require electronically steered flat-panel antennas mounted on the fuselage due to their frequent handovers. However, many business jets, being smaller than commercial aircrafts, lack the space required for these flat-panel antennas, creating a critical dependency on GEO satellites for maintaining connectivity [3]. Achieving optimal performance in these systems remains a challenging

task due to the intricate interplay of various factors. These include dynamic flight paths, varying satellite beam availability, and environmental variables such as weather conditions and seasonal fluctuations. Additionally, technical constraints like signal-to-noise ratio (SNR), network congestion, and the unpredictable nature of user demands introduce further complexities. For instance, changes in flight maneuvers or a sudden spike in user activity can significantly affect the network's ability to maintain high performance. Addressing these challenges requires advanced predictive and optimization models capable of managing the intricate interplay of these variables in real-time. Traditional approaches, such as rule-based or heuristic methods, often fall short when applied to dynamic and large-scale networks. As a result, there is a growing interest in leveraging advanced machine learning models, particularly those capable of capturing temporal and spatial dependencies, to enhance QoS and decision-making in satellite-based communication systems.

In our prior work in [4], we successfully introduced scoring and clustering algorithms to classify QoS, along with K-nearest neighbor (KNN) and Long-Short Term Memory (LSTM) models for prediction. Recognizing the limitations of these methods in capturing complex, long-range dependencies, we sought to improve prediction accuracy. This paper extends that work by exploring the application of transformer models to predict QoS in GEO satellite-based avionic communication systems. By leveraging the transformer's ability to process sequential data more effectively and in parallel, our models, trained on real-world business aviation data, demonstrate a notable improvement in performance, leading to more reliable and scalable solutions.

II. BACKGROUND

Despite the critical role of QoS prediction in aviation, existing methodologies remain limited, largely due to the inherent complexity and variability of satellite-based communication systems. Machine learning (ML) techniques present a promising solution by enabling the modeling of complex dependencies, identifying patterns and trends, and adapting to rapidly evolving conditions. Currently, no research specifically applies ML-based techniques to optimize satellite-based communication systems from the perspective of the aircraft terminal. To address this gap, we focus on prior work in satellite communication from the satellite perspective, establishing a foundation for innovative solutions and identifying the research gap. This review explores ML-based solutions to challenges in satellite communication systems and sets the stage for adapting these concepts to avionic communication. Furthermore, we present our previous work in this domain, outlining the challenges encountered and demonstrating how the proposed solutions in this paper effectively address them.

A. APPLICATION OF ML IN SATELLITE BASED COMMUNICATION SYSTEMS

We begin our review with the foundational work of [5], which offers a comprehensive overview of artificial intelligence (AI)

applications in satellite communication. This study established that various AI methodologies, including deep learning and reinforcement learning, can address critical challenges in satellite networks such as resource management, interference mitigation and traffic forecasting. Building on this, other studies have applied AI and ML to specific problems such as enhancing handover management in hybrid networks with deep learning [6] and interference management using deep learning frameworks [7].

More recent work has focused specifically on GEO constellations, addressing unique challenges from the satellite operator's perspective. For instance, studies by [8] and [9] use reinforcement learning and regression models to optimize RRM and flexible payload configurations for multi beam GEO systems, improving efficiency in dynamic traffic scenarios. Additionally, [10] presents a framework for optimizing power allocation during congestion using a supervised learning approach, demonstrating the potential of ML to enhance throughput and user satisfaction in real-time.

These studies consistently demonstrate the effectiveness of ML in improving QoS and efficiency. However, a key limitation across this body of work is its focus on optimizing the network from the satellite operator's perspective, rather than from the perspective of the user, namely an aircraft terminal. This distinction is crucial as the aircraft terminal lacks direct control over the satellite network's resource allocation, which is the primary gap our research aims to address.

In addition to the aforementioned works, recent advancements in transformer-based architectures for temporal and spatial modeling in communication systems are noteworthy. Studies like [11] explore the application of attention mechanisms for channel predictions. These emerging approaches align closely with the objectives of the present work, which aims to leverage advanced AI techniques, particularly transformer models, to optimize QoS in satellite-based communication systems for business aviation.

B. PRIOR WORK IN ML BASED AVIONIC COMMUNICATION

The prior works discussed focused on optimizing satellite systems from the operator's perspective, a problem that is distinct from the challenges faced by an aircraft terminal. Unlike ground-based users, an aircraft operates in a highly dynamic environment with constant changes in satellite visibility and frequent handovers. Moreover, the aircraft's onboard communication system lacks direct control over network configurations or resource allocation. Instead, the aircraft's system must make informed decisions with the resources it has, such as selecting from multiple available networks or optimizing a flight path. Therefore, avionic communication warrants a separate investigation focused on predictive modeling from the terminal's point of view.

ML techniques present a promising solution by enabling the modeling of complex dependencies, identifying patterns and trends, and adapting to rapidly evolving conditions. In our

earlier work [12], we investigated the use of ML-based methods to predict air traffic value and user demands within satellite beams. This study demonstrated that Variational Autoencoder Generative Adversarial Networks (VAEGANs) performed well in capturing abrupt traffic fluctuations across both high- and low-demand beams. However, a key limitation of this approach was its scalability, as the model's output dimensions grew with the number of beams, posing challenges for real-world applications where satellite constellations consist of multiple satellites with various spot beams.

Subsequently, we introduced LSTM and KNN models to predict QoS along predefined flight paths, independent of the number of beams in our work in [4]. This approach addressed the scalability issue by leveraging the LSTM architecture to capture temporal dependencies and predict performance metrics for individual flight positions. In our approach, we introduced a scoring system to evaluate the performance of the network, with scores ranging from 1 to 10, where 1 indicates the poorest performance and 10 indicates the best. The model utilized positional data and network-related features as inputs and predicted the QoS score, providing actionable predictions that could assist in optimizing flight paths for improved communication performance before flights take off, such as selecting the optimal flight plan. To evaluate the performance of this approach, we benchmarked the LSTM and KNN models against a rule-based method. The rule-based approach, described in more detail in [4], uses historical averages and rules to determine the QoS score for a given flight path. Our results revealed that both the LSTM and KNN models outperformed the rule-based approach, with the KNN model demonstrating slightly higher prediction accuracy. However, the LSTM model significantly outperformed the KNN model in terms of inference speed, making it more suitable for real-time applications.

A key challenge encountered during this study was class imbalance, as the satellite network predominantly operated at high-performance levels, resulting in a disproportionate number of samples assigned the highest performance score. While both the LSTM and KNN models provided reasonable predictions for some underrepresented classes, their performance diminished for the most sparsely represented categories. In this work, we build upon these findings by proposing a transformer-based model to improve prediction accuracy, particularly for underrepresented classes, while maintaining rapid inference times. The attention mechanism in the transformer architecture is leveraged to address class imbalance by enabling the model to better focus on informative features, thereby enhancing its ability to capture subtle variations in QoS.

C. MOTIVATION AND CONTRIBUTIONS

QoS prediction is essential in aviation. Unlike traditional approaches that focus on optimizing satellite QoS from the operator's perspective, our approach prioritizes predictive modeling from the perspective of the aircraft's onboard communication system. In avionic communication systems,

the onboard communication infrastructure is sometimes designed and operated by a separate entity from the satellite provider, meaning it has no direct control over satellite resource allocation or network configurations. Furthermore, modern aircrafts may be equipped with multiple terminals supporting different satellite operators and constellations, requiring an independent optimization strategy that accounts for connectivity across multiple networks.

By focusing on predictive modeling at the user terminal level, our approach enables the onboard communication system to proactively adjust flight paths or network selections to optimize in-flight connectivity. Rather than reacting to network fluctuations mid-flight, QoS insights can be integrated into flight planning, minimizing disruptions and enhancing passenger experience. Real-time network optimization, while valuable, is constrained by onboard processing capabilities and the lack of direct control over satellite operator decisions. Additionally, changes to a flight path must be approved by Air Traffic Control (ATC), making mid flight deviations challenging. This is precisely why our approach focuses on providing actionable insights before a flight even takes off. Predicting QoS in advance allows for a more strategic approach, enabling informed decisions such as selecting a flight path, from a set of available routes, that can provide optimal QoS. This is especially relevant in business aviation, where it is possible to select a preferred flight plan for a given city pair, leveraging predicted network conditions to ensure aircrafts remain connected to the most reliable networks throughout the flight.

This predictive approach is particularly valuable for aircrafts equipped with multiple communication terminals capable of accessing different satellite networks. By knowing the predicted performance of each network along a potential flight path, the aircraft's communication system can be configured to strategically switch to the most reliable network at specific points. This level of proactive, multi-network optimization is a powerful tool for enhancing connectivity and is a significant advantage over methods that can only react to network fluctuations as they occur.

Yet, existing approaches are predominantly reactive, addressing issues as they arise rather than proactively anticipating and mitigating the variability of satellite-based communication systems. Satellite-based avionic communication systems are characterized by dynamic factors such as high mobility, limited onboard resources, frequent handovers, and fluctuating user demands. While machine learning has shown promise in tackling similar challenges in satellite communication [5], [6], [7], [8], [9], [10], no prior research has focused on applying ML techniques from the aircraft terminal's perspective, highlighting a significant research gap.

In this paper, we address this gap by proposing a transformer-based model for predicting QoS in avionic communication systems, using real-world avionic data. Unlike the previously mentioned studies, [9], [12] [8], [10], that focus on optimizing satellite beam configurations for GEO operators, our work takes a different approach by focusing

on the needs of the aircrafts rather than satellite operators. This distinction is important, as optimizing satellite beams assumes direct control over the satellite operators, whereas our approach operates under the assumption that aircraft's communication system does not have such control. Instead, we focus on making informed predictions that can assist in flight planning decisions to enhance communication performance before flights take off, such as selecting the optimal flight path based on expected network conditions. In current industry practices, beam optimization can be highly inefficient. For example, certain customers or countries might pay very high rates to have entire beams dedicated to their fleet, even when these beams are not always in use. Our solution aims to address this inefficiency by providing more granular and dynamic predictions of QoS.

While our previous work in [4] successfully introduced a QoS prediction system using LSTM and KNN models, we noticed that their performance was limited by a significant class imbalance. This imbalance is a natural and positive characteristic of the network: it typically performs well, resulting in fewer examples of poor or average performance. Thus, this scarcity made it difficult for our previous models to accurately predict the less frequent, but critically important, instances of poor network quality. To address this challenge and continue our work, this paper introduces a suite of transformer-based models.

The proposed transformer models are designed to overcome the trade-off between speed and accuracy observed in our previous work. Our approach achieves higher overall accuracy than both the LSTM and KNN models, while simultaneously maintaining low inference times. This performance boost is particularly evident in our ability to improve predictions for underrepresented classes. By leveraging the attention mechanism, the transformer architecture can better identify subtle variations and focus on critical features in the data, thereby improving the accuracy of those less frequent, but crucial, network events. Ultimately, this work introduces a more robust and scalable solution, paving the way for more adaptive avionic communication systems. The contributions of our paper are summarized as follows:

- This paper introduces a transformer-based framework for QoS prediction in satellite-based avionic communication systems. By applying these advanced models, we directly address the limitations of our previous work [4] and significantly improve prediction accuracy, particularly for the rare, underrepresented classes
- We present a comprehensive analysis and comparison of several transformer architectures (decoder-only, encoder-decoder, and hybrid models) against established baselines (LSTM and KNN from our prior work, and a rule-based approach [4]). This rigorous evaluation quantifies the performance gains of our proposed models in terms of accuracy, RMSE, and inference speed. From this point onward, we will refer to the encoder-decoder transformer model as the proposed transformer model.

- Our work utilizes a real-world dataset comprising of positional and QoS aircraft data for flights to and from the United States of America, which is used for training and testing. This demonstrates the viability of our solution for operational use in business aviation

The remainder of this paper is structured as follows. Section III introduces the system model, outlines the problem formulation, and describes the dataset used. Section IV details the proposed models and the training process. In Section V, we assess the performance of the proposed models across various metrics and compare them against LSTM and KNN as benchmarks. Finally, Section VI presents concluding remarks.

III. SYSTEM AND PROBLEM FORMULATION

In this section, a GEO-based avionic satellite communication system for in-flight connectivity in business jets is described. Next, the QoS prediction problem is described. Finally, we discuss the dataset used as well as the steps taken to process it.

A. SYSTEM MODEL

This study focuses on a GEO satellite-based avionic communication system designed for business jets. The system enables in-flight connectivity by establishing a communication link between the aircraft and a ground station via GEO satellites, as shown in Figure 1. The aircraft communication system is composed of 3 main components.

- **Aircraft Antenna:** Mounted at the tail of the aircraft, this component is responsible for establishing a line-of-sight connection with the satellite. The antenna dynamically adjusts its orientation to maintain optimal communication with the target satellite or beam. We consider a system with a mechanically steered antenna [13]
- **Modem:** Processes the signal received by the antenna and interfaces with the onboard networking system. It also plays a critical role in ensuring compatibility with multiple satellite beams and facilitates beam switching during handovers [13]
- **Router:** Manages onboard network traffic and provides internet access to users. The router optimizes data flow based on the available bandwidth and prioritizes certain traffic types to ensure reliable connectivity [13]

The GEO satellite constellation consists of various satellites. These satellites are a combination of high-throughput satellites (HTS) with spot beams and wide-beam satellites. Spot beams offer high capacity and localized coverage, while wide beams provide broader coverage with reduced capacity [14]. Each beam is characterized by its maximum information rate, capacity, signal-to-noise ratio (SNR) shown in Equation 1, and polarity. As the aircraft traverses its flight path, it dynamically connects to different satellites and beams. The selection of satellites and beams is influenced by multiple factors, including flight path, beam capacity, user demand and SNR. The flight path determines the geographic



FIGURE 1. Diagram describing a satellite-based avionic communication system.

availability of beams and satellites, the beam capacity ensures adequate resources are available to handle user demand which varies based on the number of users on board and their bandwidth requirements and SNR affects the quality and reliability of the connections.

$$SNR = 10\log_{10}\left(\frac{P_{signal}}{P_{noise}}\right) \quad (1)$$

The system also handles various types of handovers, which could impact the handover time differently. For instance, satellite handovers generally take more time to complete compared to beam handovers. Break-before-make (BBM) beam handovers, which often involve a switch between beams with opposite polarizations, require even longer transition periods [15]. In a BBM handover, the current connection is terminated before a new one is established. This typically occurs when switching between beams with opposite polarizations or when transitioning between beams with non-overlapping coverage areas. In such cases, maintaining both connections simultaneously is technically infeasible, necessitating the disconnection of the existing link before establishing the new one [16]. The total handover time is shown in Equation 2 and is described as the difference in time between when a new beam is acquired and a handover is attempted. In our system model, handover decisions are primarily managed by the network provider's ground-based operations center. However, the aircraft's onboard system continuously monitors real-time signal metrics, such as SNR and throughput. If these metrics drop below a predefined threshold, the onboard system can request a handover. The final decision and execution of the handover, however, remain under the control of the network's operations center [17].

$$t_{handover} = t_{acquisition} - t_{attempt} \quad (2)$$

To take these variable factors into consideration, we can analyze the user's throughput to assess the performance of their in-flight connectivity, as throughput reflects how effectively the network resources are being allocated and utilized under varying conditions. First, we look at Shannon's capacity in Equation 3 which describes the total achievable rate C given the bandwidth B and SNR [18]. This equation provides the maximum theoretical data rate that can be achieved over a communication channel, accounting for both the available bandwidth and the quality of the signal [18].

$$C_{shannon} = B\log_2(1 + SNR) \quad (3)$$

The maximum achievable user throughput at any given time, shown in Equation 4, is determined by the minimum of Shannon's capacity, $C_{shannon}$, and the ratio of the beam's maximum information rate (MIR), C_{max} , over $1 + \gamma \cdot Congestion$. In this formula, γ represents a variable indicating the intensity of congestion, while $Congestion$ is a binary flag that takes the value of 1 if the beam is congested. If $Congestion$ is 0, the ratio equals C_{max} , meaning the user can receive the beam's full MIR. Otherwise, the maximum throughput is limited by the system's SNR and bandwidth as described by Shannon's capacity.

$$T_{max} = \min(C_{shannon}, \frac{C_{max}}{1 + \gamma \cdot Congestion}) \quad (4)$$

B. PROBLEM FORMULATION

To assess the QoS on the aircraft and incorporate all the factors described above, we use a QoS score derived from a K-means clustering algorithm, which was first introduced in our work in [4]. This score considers two key factors: the downstream allocation relative to demand and the signal-to-noise ratio (SNR). The allocation to demand is defined as the ratio of downstream allocated throughput to demanded throughput, as shown in Equation 5. In this context, the maximum demand a user can request is the beam's MIR, C_{max} , while the maximum throughput a beam can allocate to a user is T_{max} , as shown in Equation 4. The MIR is the highest data transmission rate that a user can achieve under optimal network conditions. This represents the peak bandwidth available to a user when the network is not congested [19]. Therefore, if the user is requesting the maximum allowable throughput then the allocation to demand, α , reaches its maximum value of 1 when $T_{allocated} = T_{max} = C_{max}$.

$$\alpha = \frac{T_{allocated}}{T_{demand}} \quad (5)$$

The SNR plays a critical role in determining the system's operability. If the SNR falls below a certain threshold, the system is unable to establish or maintain a reliable communication link [20]. This occurs because low SNR values indicate that the signal is either overwhelmed by noise or experiencing degradation, which could result from flight maneuvers, antenna blockage, weather conditions or equipment failures. These conditions lead to reduced data transmission quality,

increased packet loss, and higher latency, ultimately rendering the connection unusable.

For terminals that remain operational, the QoS score is primarily influenced by the allocation of resources to meet user demand. Higher allocations relative to demand translate to better QoS, as more resources are available to support user activities without congestion or service degradation. The allocation to demand is mapped to a score using k-means. The clustering algorithm groups data based on the similarity of the downstream allocation to demand by partitioning the data into K clusters by minimizing the variance within a cluster. The objective is to find cluster centroids that best represent the data points within each group. This can be formalized by the optimization problem shown in Equation 6, where α_i is the sample of allocation to demand and μ_k is the centroid of the cluster C_k and $1_{\{x_i \in C_k\}}$ is an indicator function that equals 1 if x_i belongs to cluster C_k and 0 otherwise.

$$\argmin_{\{C_1, C_2, \dots, C_K\}} \sum_{i=1}^n \sum_{k=1}^K 1_{\{x_i \in C_k\}} \|\alpha_i - \mu_k\|^2 \quad (6)$$

This scoring system enables an evaluation of the communication system's effectiveness under varying conditions and demands. The use of a clustering algorithm allows us to contextualize the user's experience relative to the dataset by grouping similar allocation-to-demand ratios into clusters ranked by allocation to demand. This ranking provides a clear interpretation of performance levels across the system, facilitating a more intuitive understanding of how resource allocations impact user experience under different operational conditions. As shown in Equation 7, for each sample, if the SNR exceeds the threshold, the score is assigned the rank of the cluster C_j to which the downstream allocation to demand, α belongs. Otherwise, the score is assigned a value of 1, indicating the terminal is not operating due to low SNR.

$$S_i = \begin{cases} R(C_j), & \text{if } \text{SNR} > \text{Threshold}_{\text{SNR}} \text{ and } \alpha_i \in C_j, \\ 1, & \text{otherwise.} \end{cases} \quad (7)$$

The problem then becomes to develop a model or function that predicts the QoS score S_i for a given input X_i , where X is a matrix of input features. Additionally, we extend the problem to a sequence to sequence prediction model, where S is a sequence of scores $\{S_1, S_2, \dots, S_T\}$, representing different timestamps within a flight. Correspondingly, X is a sequence of inputs $\{X_1, X_2, \dots, X_T\}$, where each input X_t is a vector of features, $[x_1, x_2, \dots, x_n]$, describing the state of the system at a given timestamp t . This formulation accounts for how network conditions and resource allocations evolve over the course of the flight. The goal is to learn a mapping, f , such that $S = f(X)$, where f predicts the sequence of scores S given the sequence of inputs X .

We formulate the problem as a classification task rather than a regression task. Regression assumes continuous output and penalizes squared differences between predictions and true values. Here, even small deviations from the true

score contribute to the loss. On the other hand, classification loss focuses on the probabilities assigned to discrete classes, which measures how well the model predicts the correct class and ignores small deviations within the same class. Hence, instead of predicting the QoS score S_t as a continuous value, the model predicts a one-hot encoded vector representing the discrete score $\hat{S}_t \in \mathbb{R}^{10}$. Each element in $\hat{S}_t$ corresponds to a class, where the class labels range from 1 to 10, ensuring that the predicted score aligns with the clustered ranking system. Thus, we design a model f_θ , parametrized by θ , that maps the input feature sequence X to the output score sequence S . The objective is to optimize the model parameters, θ , such that the loss function $\mathcal{L}(S, \hat{S})$, which quantifies the discrepancy between the predicted scores $\hat{S}$ and the true scores S , is minimized as shown in Equation 8.

$$\theta^* = \argmin_{\theta} \mathcal{L}(S, \hat{S}) \quad (8)$$

C. DATA PREPARATION

For ease of comparison, we take the same data preparation steps taken in our earlier work in [4]. We utilize a real-world dataset containing anonymized positional and QoS information from a GEO satellite constellation operating in the Ku-band for business aviation jets. The dataset encompasses timestamps, positional coordinates, satellite and beam identifiers, handover probabilities, historical averages for SNR and maximum information rate (MIR), as well as the user's historical average flight performance and usage patterns, which makes up the features of the input sequence X . To determine the average historical SNR and MIR, we segment the beam footprint into smaller hexagonal regions, as shown in 2. The historical SNR and MIR are then computed within each of these geographical regions. We consider flights to and from the United States between January to October 2024.

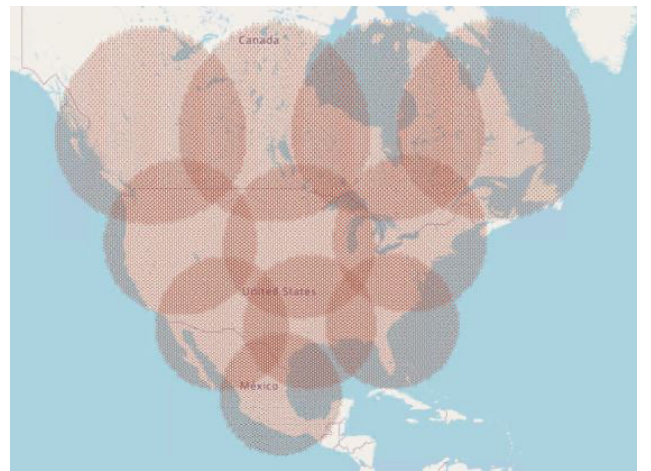


FIGURE 2. Example of satellite beam footprint separated into smaller geographical regions.

To determine the QoS score, we utilize the demanded and allocated throughput at each timestep, as provided by equipment logs, and compute the allocation to demand using

Equation 5. Subsequently, we apply the K-means clustering algorithm described in Equation 6 to categorize the downstream allocation to demand, and compute the score for each timestamp based on the downstream allocation to demand and SNR using Equation 7. To estimate the handover probability, we employ the same DBScan based clustering algorithm introduced in our work in [4] which calculates the likelihood of a satellite, break before make and make before break handovers across varying geographic locations within the beam. This approach accounts for aircraft headings in the North West, North East, South West, and South East directions.

The dataset is sampled at 10-minute intervals, with time and position data augmented to include season, weekend, and holiday indices, as well as sine and cosine transformations of longitude and latitude. To further enhance our model's robustness and generalization, we performed additional data augmentation on the training set by randomly flipping sequences and adding small amounts of noise to the input features. The input features are as follows: year, month, day, hour, minute, season, holiday, and weekend indices, aircraft tail number, destination and departure airports, sine and cosine of longitude and latitude, altitude, heading, distance and time since the start of the flight, distance and time to the destination, historical averages for SNR, MIR, score, and number of connected devices, satellite and beam names, and probabilities for make-before-break and break-before-make handovers. Notably, the input features are selected to ensure availability before the flight, excluding any real-time performance indicators. This restriction aligns with the model's purpose of performing pre-flight network performance predictions.

We treat each flight as a sequence where the input features make up the sequence X as shown in Equation 9 with the columns representing features and the rows representing timestamps within the flight. The output sequence S is the one-hot encoded value of the score, as shown in Equation 9, where each column represents a different output class, and the rows represent the corresponding timestamps.

$$\begin{bmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,n} \\ x_{2,1} & x_{2,2} & \dots & x_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{T,1} & x_{T,2} & \dots & x_{T,n} \end{bmatrix} \begin{bmatrix} s_{1,1} & s_{1,2} & \dots & s_{1,10} \\ s_{2,1} & s_{2,2} & \dots & s_{2,10} \\ \vdots & \vdots & \ddots & \vdots \\ s_{T,1} & s_{T,2} & \dots & s_{T,10} \end{bmatrix} \quad (9)$$

where:

- $x_{t,1}, x_{t,2}, \dots, x_{t,n}$ are the input features for a flight,
- $s_{t,1}, s_{t,2}, \dots, s_{t,10}$ are the corresponding output class probabilities for each timestamp in a flight,
- T represents the total number of timestamps within a flight.

IV. PROPOSED MODEL

In this section, we introduce the proposed transformer-based model for QoS predictions in satellite-based avionic communication systems. Specifically, we explore four distinct transformer architectures: a decoder-only transformer,

an encoder-decoder transformer, an encoder-decoder transformer with an LSTM layer, and an encoder-decoder transformer with a convolutional layer.

Transformers are a class of deep learning models originally developed for natural language processing tasks, but their adaptability and effectiveness have made them highly effective for various sequence-to-sequence prediction problems. Unlike traditional models, transformers rely on attention mechanisms rather than recurrent connections, enabling them to process sequences in parallel. This parallelism significantly reduces training time and makes transformers highly scalable to large datasets [21]. For sequence-to-sequence tasks, such as predicting QoS in satellite-based avionic communication systems, transformers excel due to their ability to capture long-range dependencies in sequences through attention mechanisms and adapt to varying input and output sequence lengths.

The proposed models will be used for offline inference, meaning they will not run onboard the aircraft and are not subject to real-time constraints. Instead, the predictions will take place before the flight takes off, allowing sufficient time for decision-making processes that can enhance in-flight connectivity. This pre-flight analysis can be leveraged to select an optimal flight path, reducing the likelihood of connectivity disruptions and improving overall QoS.

Before introducing the model architectures, we will first discuss the functions of the mechanisms outlined below.

A. MECHANISMS

1) INPUT EMBEDDING

The input embedding layer transforms our input feature matrix X , shown in Equation 9, into a dense vector representation of fixed dimensionality [21]. Input embeddings map the input feature matrix X from a dimension of TXn to a dimension of TXd , where T is the length of the sequence, n is the dimensionality of the input features and d is the dimensionality of the hidden space. This transformation allows the model to process the data in a high-dimensional space, enabling it to learn relationships and patterns effectively. In this section, we will refer to the output of the input embedding layer as $\hat{X}$, where $X \xrightarrow{\text{InputEmbedding}} \hat{X}$.

2) POSITIONAL ENCODING

Transformers lack inherent order information, as they process inputs in parallel. Positional encoding introduces order by adding fixed or learned positional information to the input embeddings [21]. In our proposed models, positional encoding is used to represent the sequence's temporal structure, such as the order of positional data or timestamps, ensuring the model can distinguish between inputs occurring at different points. To implement positional encoding we first generate a positional encoding matrix, PE , is shown in Equation 10. The position, pos , is extracted from the embedded input matrix, $\hat{X}$, where each element in the matrix

represents features of dimensionality d .

$$\begin{bmatrix} \hat{X}_1 \\ \hat{X}_2 \\ \vdots \\ \hat{X}_T \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 2 \\ \vdots \\ T \end{bmatrix} \rightarrow \begin{bmatrix} PE_{1,1} & PE_{1,2} & \dots & PE_{1,d} \\ PE_{2,1} & PE_{2,2} & \dots & PE_{2,d} \\ \vdots & \vdots & \ddots & \vdots \\ PE_{T,1} & PE_{T,2} & \dots & PE_{T,d} \end{bmatrix} \quad (10)$$

We generate the matrix using Equation 11 and Equation 12 [21]. This ensures that each position in the sequence, pos , is mapped to a unique vector, with alternating sine and cosine values for even and odd dimensions. The periodicity of these functions enables the model to generalize positional relationships.

$$PE_{pos,2i} = \sin\left(\frac{pos}{\rho^{\frac{2i}{d}}}\right) \quad (11)$$

$$PE_{pos,2i+1} = \cos\left(\frac{pos}{\rho^{\frac{2i}{d}}}\right) \quad (12)$$

where:

- pos is the position in the sequence
- i is the dimension index
- d is the dimensionality of the embedding space
- ρ is a scaling factor ensuring a smooth distribution of the encoding values

Next, we add the positional encoding matrix to the input matrix X as shown in Equation 13. This step integrates positional information into the input embeddings.

$$X_{output} = \hat{X} + PE \quad (13)$$

3) ATTENTION

The attention mechanism is a fundamental concept in transformers as it allows the model to focus on specific parts of the input sequence [21]. The core idea is to calculate a weighted sum of the input values where the weights are dynamically determined based on the relationships between the tokens in the sequence. This enables the models to selectively attend to different parts of the input, capturing relevant dependencies between tokens regardless of their position in the sequence.

The query, key and value are linear transformation of the encoded representation of the input sequence, X_{output} . They are represented as $Q = X_{output}W_Q$, $K = X_{output}W_K$ and $V = X_{output}W_V$, where Q , K and V are the query, key and value vectors and W_Q , W_K and W_V are learnable weight matrices. The query represents what the model is searching for, the key represents what each input element has to offer, and the value contains the actual information to be attended to and passed along. The attention weights are then computed using Equation 14 [21]. The context vector is the weighted sum of the value vectors and attention weights, $\alpha_w V$. This vector encapsulates the most relevant information from the input sequence for a given query, summarizing the relationship and dependencies across the sequence.

$$\alpha_w = \text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) \quad (14)$$

In self-attention, the query, key and value are all derived from the same input sequence [21]. This allows the model to compute relationships between tokens within the same sequence. Whereas in cross-attention, the query vector is derived from one sequence while the key and value vectors are derived from another sequence [21]. This is particularly useful when the model needs to align information between two different sequences. Additionally, multi-head attention extends the attention mechanism by splitting the query, key and value vectors into multiple smaller subspaces, referred to as heads [21]. Each head independently computes its own attention output, allowing the model to attend to different aspects of the sequence simultaneously. The outputs from all the heads are then concatenated and linearly transformed to produce the final output. Multi-head attention allows the model to capture a richer set of relationships within or between sequences, as each head focuses on a unique subset of features.

B. PROPOSED MODELS

We propose a decoder-only transformer, an encoder-decoder transformer, an encoder-decoder transformer with LSTM and an encoder-decoder transformer with convolution. The architecture of the proposed models is discussed below.

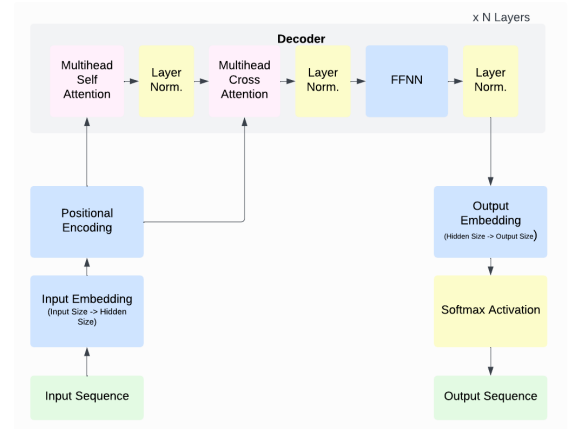


FIGURE 3. Decoder-only transformer model architecture.

1) DECODER-ONLY TRANSFORMER

The decoder-only transformer model, shown in Figure 3, consists of an input embedding layer, a positional encoding layer, a multi-head self-attention mechanism, a multi-head cross-attention mechanism, a feedforward layer, layer normalization, an output layer, and a softmax activation applied to the output.

Layer normalization is a technique used to improve the stability and performance of deep neural networks. It normalizes the activations of a layer across its features for each input example, ensuring that the input to a layer has a mean of

0 and a variance of 1 [22]. This normalization helps improve convergence and training stability.

The model takes the input feature matrix X , which represents the encoded input sequence, and outputs class probabilities representing the QoS score S , as described in Equation 9. The decoder processes a full sequence as input and produces a full sequence as output, with each sequence corresponding to a flight. To achieve this, we propose a non-autoregressive decoder that does not rely on previous outputs, enabling faster and parallelized decoding.

As shown in Figure 3, the self-attention mechanism is applied to the encoded representation of the input sequence, enabling the model to capture relationships within the sequence. The cross-attention mechanism is then performed on the output of the self-attention mechanism and the encoded representation of the input sequence, allowing the decoder to align and integrate information from the input.

The model parameters are summarized below:

- Number of Layers: 4
- Number of Attention Heads: 8
- Hidden Dimension, d : 128
- Input dimension, n : 36
- Output Dimension: 10

The selection of these hyperparameters was determined through an iterative process of experimentation. We started with smaller values for the number of layers and attention heads and progressively increased them. Our findings indicated that using four layers and eight attention heads provided the optimal balance between model complexity and performance. While a higher number of layers and heads initially improved accuracy, the gains eventually plateaued, and further increases resulted in diminishing returns and higher computational costs. This configuration was found to be the most effective for our dataset, delivering high predictive accuracy without unnecessary overhead.

2) ENCODER-DECODER TRANSFORMERS

The encoder-decoder transformer model, shown in Figure 4, consists of an encoder and a decoder. The encoder processes the input sequence to produce a sequence of context-rich representations, while the decoder generates the output sequence based on both the original input and the encoder's output.

The encoder consists of an input embedding layer, a positional encoding layer, multi-head self-attention mechanisms, and feedforward layers. The attention blocks are followed by feedforward layers and layer normalization to improve stability and training efficiency.

We use the same decoder architecture as the decoder-only transformer. The decoder receives as input the concatenation of the original signal and the output of the encoder. This combination allows the decoder to leverage both the encoded sequence and the raw input, providing a richer context for generating the output sequence. Like the encoder, the decoder applies a multi-head self-attention mechanism to the concatenated input, followed by a cross-attention mechanism that enables the decoder to align its output with the encoder's

output. This allows the model to capture both local and global dependencies within the input sequence, as well as dependencies between the input and output sequences.

Similar to the decoder-only transformer, the self-attention mechanism is applied to the encoded representation of the input sequence, enabling the model to capture relationships within the sequence. The cross-attention mechanism is then applied to the output of the self-attention mechanism and the encoded representation of the input sequence, allowing the decoder to align and integrate information from the input.

In addition to the standard encoder-decoder transformer, we propose two alternative encoder-decoder transformer models, each incorporating an LSTM or a convolutional layer at the input. As shown in Figure 4, the architecture of these models remains the same as the standard transformer model, with the exception that an LSTM or convolutional layer is added to the input layer, as indicated by the dashed block in the figure.

The inclusion of both an LSTM and a convolutional layer in the hybrid models serves as a form of preprocessing before the input data reaches the transformer's encoder. These layers act as feature extractors, aiming to provide the transformer's global attention mechanism with a richer, more refined representation of the input data.

The LSTM layer is used to capture long-term temporal dependencies within the sequential input data. While the transformer's attention mechanism is capable of modeling these relationships, the LSTM provides an alternative method to explicitly learn and represent the sequential flow of information. The hypothesis was that this might provide the transformer with a more robust understanding of the temporal context, potentially improving performance on complex sequences.

Similarly, the convolutional layer is used to extract local features from the input sequence. By applying a sliding window filter, the CNN can identify patterns and correlations across neighboring data points. This is particularly useful for tasks where the local context is important. The hope was that this preliminary feature extraction would complement the transformer's global attention, which looks at the entire sequence simultaneously, by providing it with a set of well-defined local features to consider.

We use the same model parameters as the decoder-only model, referenced above.

C. MODEL TRAINING

For training, we utilize the Adam optimizer with a weight decay factor of 0.0001 to regularize the model and mitigate overfitting. The initial learning rate is set to 0.001, and a Reduce-on-Plateau learning rate scheduler is employed. This scheduler reduces the learning rate by 10% whenever the validation loss fails to improve for 5 consecutive epochs, ensuring a smooth and adaptive optimization process.

The model is trained with a batch size of 128, and gradient accumulation is implemented with an accumulation factor of 4, resulting in an effective batch size of 512. Gradient

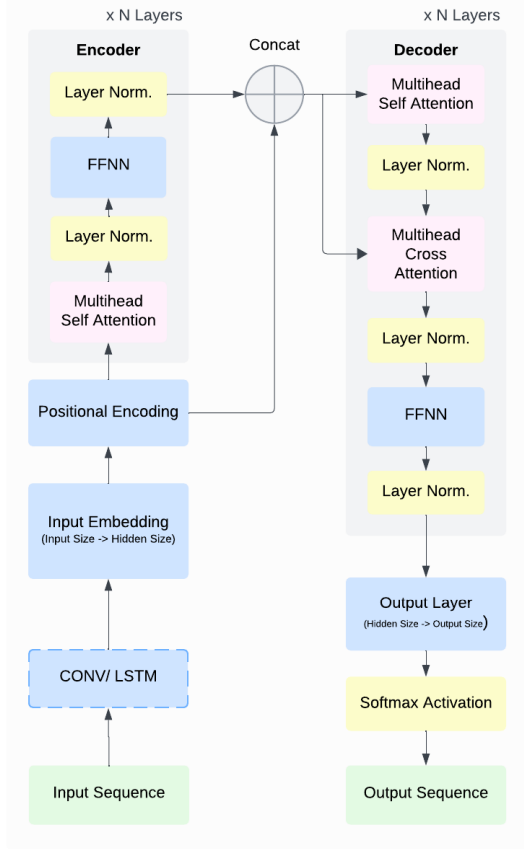


FIGURE 4. Encoder-Decoder Transformer model architecture.

accumulation allows the summation of gradients over multiple mini-batches before performing an optimization step, enabling stable training while remaining within memory constraints [23]. This setup promotes efficient convergence without compromising computational feasibility. We train the model for an average of 200 epochs. To further enhance training efficiency, we apply early stopping by monitoring the validation loss. Training is halted when the validation loss stops decreasing for 5 consecutive epochs, ensuring the model does not overfit and is trained for an optimal number of epochs.

We use PyTorch [24] to implement and evaluate our models, leveraging its flexibility and extensive library support for deep learning tasks. For the attention mechanism, we utilize the FlashAttention library [25] to significantly accelerate training. FlashAttention optimizes the computation of attention scores, reducing memory overhead and enabling faster training without compromising accuracy.

We use a weighted cross-entropy loss, shown in Equation 15, to address the challenge of class imbalance in our dataset. In this context, class imbalance arises because the network typically performs well, leading to a significantly larger number of samples with high scores compared to those

with low scores. As a result, the model might become biased toward predicting high-score classes, neglecting low-score predictions that are critical for comprehensive evaluation.

To tackle this issue, the weighted cross-entropy loss assigns higher penalties to low-score classes, ensuring that the model effectively learns to predict these less-represented classes. By applying weights inversely proportional to the frequency or significance of the classes, the loss function encourages the model to pay greater attention to low-score classes during training. The weights are inversely proportional to the class index, assigning higher weights to lower score classes. This weighting scheme ensures that low-score classes receive increased importance during the optimization process. We experimented with several loss functions, including cross-entropy and focal loss, but found that weighted cross-entropy provided the best overall performance. This approach allowed us to assign a higher penalty to under-represented, low-score classes while still maintaining strong prediction accuracy for the most frequent class, Class 10. By strategically tuning the weights, we were able to minimize false positives in Class 10, preventing unnecessary alarms for customers, while also ensuring high accuracy for Class 1 to effectively detect critical anomalies.

$$\mathcal{L}_{\text{weighted}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C w_c \cdot y_{i,c} \cdot \log(\hat{y}_{i,c}) \quad (15)$$

where:

- N is the total number of samples
- C is the total number of classes
- w_c is the weight assigned to class c
- $y_{i,c}$ is the one-hot encoded ground truth for class c of sample i
- $\log(\hat{y}_{i,c})$ is the logarithm of the predicted probability

V. PERFORMANCE EVALUATION

We evaluate the performance of our proposed transformer models by comparing them against each other, as well as against two baselines, namely the LSTM and KNN models from our work in [4]. Additionally, we include a non-ML rule-based approach as a baseline for comparison. This comprehensive evaluation enables us to assess the effectiveness of the proposed models in various aspects of prediction and classification.

Our evaluation framework uses multiple metrics to provide a comprehensive assessment of our models. We compute standard classification metrics, such as precision recall, F1 score, and accuracy, as shown in Figure 1 to evaluate how well the models classify instances into their correct categories. However, accuracy alone can be misleading in our case. It treats a prediction of 1 for a true value of 10 the same as a prediction of 9 for a true value of 10, even though the latter is a much better estimate from an operational perspective. Because accuracy only measures whether the prediction is exactly right, it fails to account for how close a prediction is to the true value. Therefore, we also use the Root Mean Squared

TABLE 1. This table summarizes the performance of each model. The abbreviations represent: P (Precision), R (Recall), F1 (F1-Score), A (Accuracy), RMSE (Root Mean Squared Error), and STD (Standard Deviation).

Model	P	R	F1	A	RMSE	STD
Dec.-only Transformer	46	63	63	63	1.99	1.96
Enc.-Dec. Transformer	66	65	65	65	1.9	1.96
Enc.-Dec. Transformer with Focal Loss	59.6	56.9	57	56.9	2.23	2.19
Enc.-Dec. Transformer with LSTM	65	65	65	65	1.9	1.87
Enc.-Dec. Transformer with CONV	66	65	65	65	1.91	1.86
LSTM	57.6	61.0	57.3	61.0	2.34	2.32
KNN	62.3	63.8	62.7	63.8	1.87	1.87
Rule-Based	51.8	7.56	3.01	7.56	2.48	2.08

Error (RMSE). By mapping our predicted class probabilities to an integer between 1 and 10, we can treat the output as an ordinal value and measure how far our predictions are from the true value. This metric helps to quantify the magnitude of our prediction errors. Yet, even RMSE can be misleading. A naive model that always predicts Class 10 might have an RMSE lower than our best-performing model because it correctly predicts the most frequent class with high confidence. This highlights the core challenge: we need a model that not only performs well on the majority class but is also capable of accurately identifying anomalies or instances of poor performance. By using a combination of these metrics, along with confusion matrices, we can ensure our evaluation is robust and provides a complete picture of model performance, demonstrating that our best model effectively balances accuracy on the majority class with the ability to detect crucial anomalies.

Table 1 displays the precision, recall, F1-Score, accuracy and RMSE of the models. The results show that all ML-based models outperform the non-ML rule-based approach. As noted in our previous work in [4], the KNN model outperformed the LSTM model. However, by leveraging the more advanced transformer-based architectures, we observe substantial performance improvements.

While the KNN model achieves the lowest RMSE of 1.87, this metric alone is misleading due to the dataset's bias towards class 10. This is evident by observing the overall accuracy, where the encoder-decoder based transformer models achieves 65% accuracy, outperforming the KNN model at 63.8%.

A deeper analysis of the confusion matrices in Figure 9 reveals the full picture. These matrices provide detailed

insights into class-wise predictions, including true positives, true negatives, false positives, and false negatives. They help identify classes where the model performs well and highlight those where it struggles.

We start by observing Figure 9a, which presents the confusion matrix for the rule-based approach, and shows that predictions are heavily concentrated around labels 8 and 9, with poor performance in correctly classifying other classes. This suggests that the rule-based approach lacks flexibility and may be biased toward these labels. Moving to the machine learning models Figures 9b and 9c illustrate the results of the KNN and LSTM models, respectively, demonstrating an improvement in predicting a wider range of classes. However, both models still exhibit a strong bias toward class 10. The decoder-only transformer, shown in Figure 9d, achieves significant improvements over the previous models, with a well-defined diagonal indicating higher accuracy. Despite this progress, some misclassifications persist, particularly in the lower classes.

The proposed transformer model, shown in Figure 9e demonstrates the highest number of true positives across all classes, showcasing its effectiveness in capturing complex patterns within the data. However, it still struggles with some underrepresented classes, particularly class 4 and class 5. Despite these challenges, the proposed transformer model shows significant improvement compared to the baselines. For instance, the percentage of true positives in class 2 increased dramatically from 9% with the KNN model and just 1% with the LSTM model to 34% with the proposed transformer model. This improvement underscores the advantages of leveraging an encoder-decoder architecture and attention mechanisms in handling diverse and imbalanced datasets.

Figures 9f and 9g illustrate the performance of the encoder-decoder transformer with LSTM and convolutional layers, respectively. Their results closely align with those of the proposed transformer model, indicating that incorporating these additional mechanisms at the input stage did not lead to a substantial performance boost.

When directly comparing the top-performing models, a closer look at their confusion matrices reveals that although the KNN model has a slightly higher true positive rate for class 10 at 83% compared to the encoder-decoder transformer's 80%, proposed transformer model's performance is notably better for the less frequent classes. Particularly, we observe the following:

- Classes 1, 2, and 3: The proposed transformer model achieves true positive rates of 89%, 34% and 83%, respectively, while the KNN model only manages 72%, 9%, 69%
- classes 4 and 5: While both model exhibit poor performance here, the proposed transformer model still outperforms the KNN model at 18% and 12% in comparison to the KNN's 11% and 11%
- Classes 6, 7, 8, 9: proposed transformer model's true positive rates, 39%, 44%, 27% and 30%, are consistently

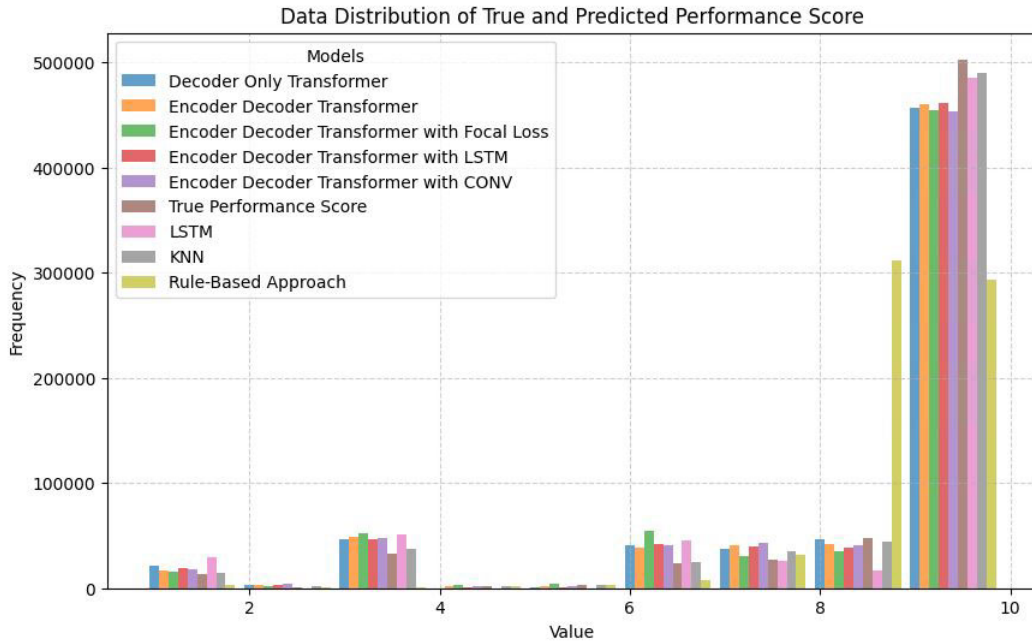


FIGURE 5. Data distribution of testing dataset and model predictions.

higher than the KNN model's which are at 20%, 29%, 22% and 23%

- The KNN achieves a higher true positive rate of 83% compared to the proposed transformer model at 80%. However, the proposed transformer model as a much lower false positive rate

We also note that the model's accuracy varies significantly across classes by observing the confusion matrices in Figure 9. This is due to the class imbalance in our dataset, where the majority of samples belong to class 10, as shown in Figure 5. This behavior is expected of the satellite communication network, as it typically tends to perform well, with fewer instances of poor performance. Our goal is to design a model that best captures this distribution.

We used weighted cross-entropy loss to address this. We found that if all classes are weighted equally, then the model would be heavily skewed towards class 10. Conversely disproportionately high weights assigned to the lower classes would cause the model to ignore the higher classes, leading to a high false negative rate particularly to class 10. The chosen weighting scheme represents a careful balance, giving the model a high true positive rate for class 10 while improving its performance on the less frequent classes. We empirically tested the performance of various loss functions and found that weighted cross-entropy with our chosen weight scheme performed the best. Our trials included a comparison with focal loss, which is often used for imbalanced datasets. As shown in Table 1, the use of focal loss resulted in a higher RMSE of 2.23 and a lower accuracy of 56.9% for the proposed transformer model. This indicates that it did not provide the same benefits in learning the underrepresented classes as our chosen weighted cross-entropy scheme. This

approach represents a careful balance, giving the model a high true positive rate for Class 10 while also improving its performance on the less frequent classes.

The particularly poor performance of Classes 2, 4, and 5 across all models is likely due to their low frequency. As shown in Figure 5, these classes are the least represented in our dataset. While our use of weighted cross-entropy loss and data augmentation helped to somewhat compensate for this imbalance, it wasn't enough to allow the models to fully learn the distinct characteristics of these classes.

Interestingly, despite their low frequency, Classes 1 and 3 achieve good accuracy across all models. A noteworthy observation is that while the LSTM and transformer-based models all use the weighted cross-entropy loss to compensate for the frequency of the least represented classes, the KNN model does not and still manages to achieve good performance for Classes 1 and 3. This is likely due to the inherent nature of the KNN algorithm, as it makes predictions based on the features of its nearest neighbors. Classes with very distinct features, such as those associated with very poor network performance, tend to form separate clusters in the feature space. As a result, the KNN model can accurately predict these classes even without a weighted loss function because their features make them easily distinguishable from other classes. Nonetheless, the transformer-based models still demonstrate better overall performance than KNN precisely because our targeted techniques provided a more robust solution for the challenging, low-frequency classes.

To assess the sequence-level similarity between the predicted and target outputs, we calculate the correlation coefficient between the true and predicted sequences, shown in eq. 16 [26]. We use the Pearson correlation coefficient to

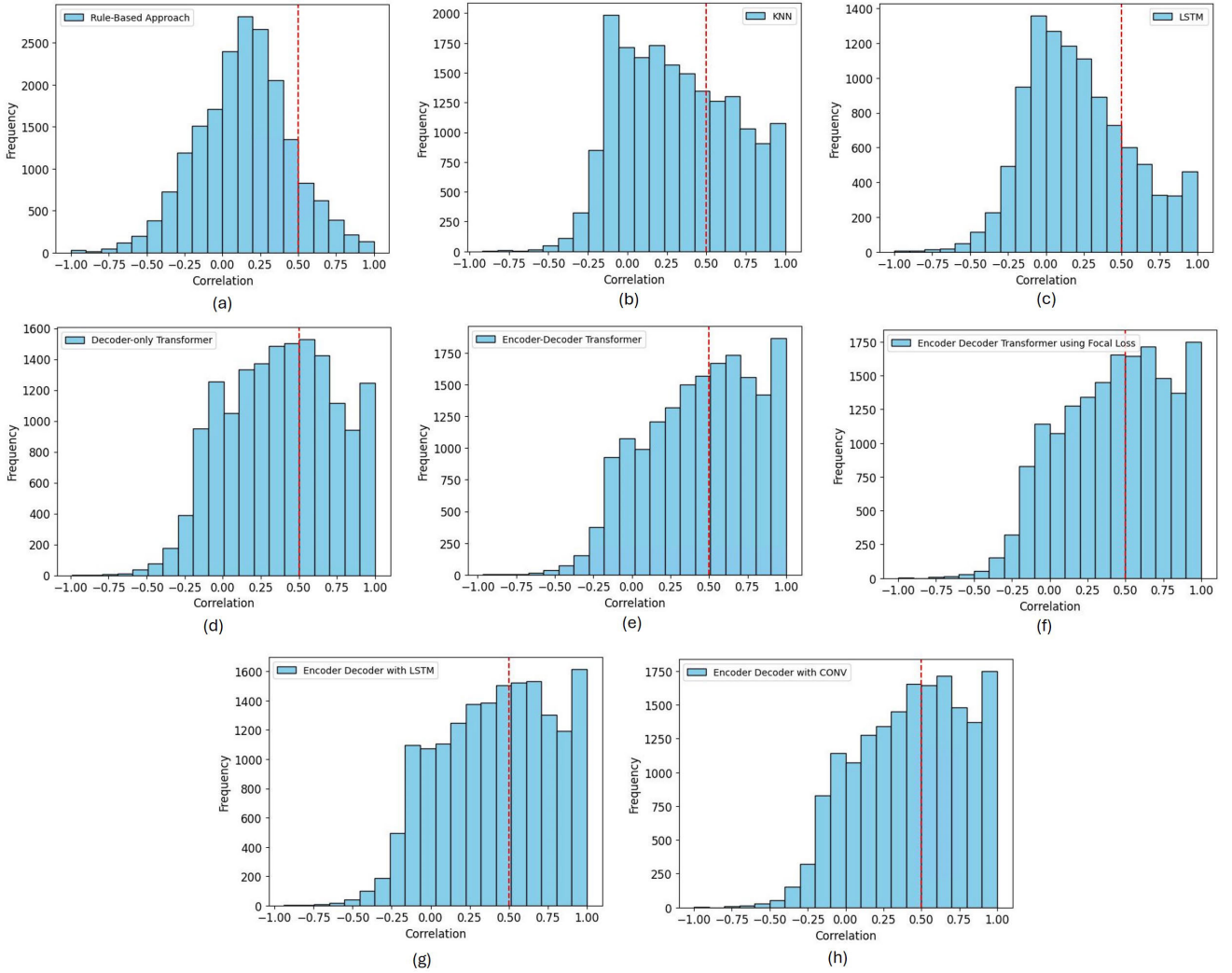


FIGURE 6. Distribution of correlation coefficients between true and predicted sequences (a) Rule-Based approach, (b) KNN Model, (c) LSTM Model, (d) Decoder-Only Transformer, (e) Encoder-Decoder Transformer, (f) Encoder-Decoder Transformer with Focal Loss (g) Encoder-Decoder Transformer with LSTM, and (h) Encoder-Decoder Transformer with Convolution Layer.

measure the relationship between the predicted and target outputs.

$$r = \frac{\sum_{i=1}^n (y_{predi} - \bar{y}_{pred})(y_{truei} - \bar{y}_{true})}{\sqrt{\sum_{i=1}^n (y_{predi} - \bar{y}_{pred})^2} \cdot \sqrt{\sum_{i=1}^n (y_{truei} - \bar{y}_{true})^2}} \quad (16)$$

where:

- r is the Pearson correlation coefficient
- y_{predi} and y_{truei} are the individual points of the predicted and target sequences
- $\bar{y}_{pred}$ and $\bar{y}_{true}$ are the means of the predicted and target sequences
- n is the number of data points in the sequence

This metric quantifies the degree of linear relationship between the sequences, offering a view of how closely the predicted sequences align with the true sequences. Unlike classification metrics or RMSE, the correlation coefficient evaluates the structural alignment between sequences,

making it particularly valuable for understanding trends and patterns. Figure 6 displays the density plots of the correlation coefficients for each model.

Starting with the rule-based approach, in Figure 6a the correlation coefficients are widely dispersed, with a substantial portion concentrated in the 0 - 0.3 range. This distribution underscores the limitations of the rule-based method, which lacks the flexibility to model complex relationships within the data. The KNN model, in Figure 6b shows a modest improvement, with a shift toward higher coefficients. However, a significant fraction of the values still falls below 0.5, indicating limited sequence-level alignment.

The LSTM model, in fig 6c, demonstrates an improvement with respect to the rule-based approach, with a higher distribution above the 0.5 mark. However, the ability of the model to capture sequence-level trends still falls below the KNN model.

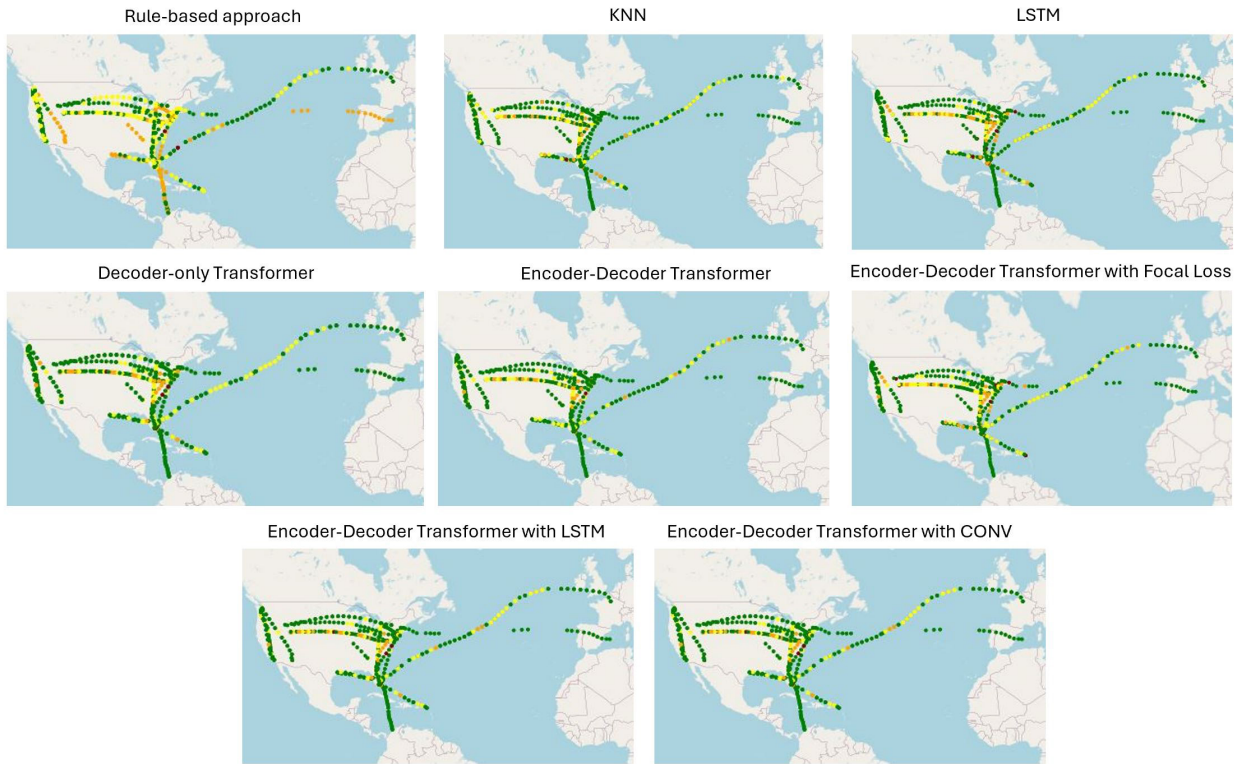


FIGURE 7. Plots visualizing the predictions of the model along sampled flight paths by categorizing the deviation of their predictions from the true values. The color of each point represents the magnitude of error. Green: $\delta \leq 1$, Yellow: $1 < \delta \leq 3$, Orange: $3 < \delta \leq 7$, Red: $\delta > 7$.

The decoder-only transformer, in Figure 16d shows a substantial shift toward higher correlation coefficients, with a notable distribution shift to the right of 0.5. This improvement suggests that the decoder-only transformer captures dependencies more effectively than both the KNN and LSTM models. The proposed transformer model, in Figure 16e achieves the most skewed distribution towards the right, with the majority of correlation coefficients exceeding 0.5. This reflects the strongest sequence-level alignment, driven by its ability to leverage bidirectional encoding and decoding mechanisms, effectively capturing both local and global dependencies in the data. In contrast, the hybrid approaches—transformer + LSTM and transformer + CONV, in Figure 16f and g, fail to outperform the standalone transformer models. This outcome underscores the sufficiency of the transformer architecture in addressing both spatial and temporal dependencies without the need for additional layers or hybrid designs.

In Table 2, we summarize model performance by categorizing the deviation of predictions from their true values. The table shows the percentage of predictions that fall into specific error ranges: less than or equal to 1 from the true value, between 1 and 3, between 3 and 7, and greater than 7. These results are particularly insightful, as they show that while the overall accuracy of the proposed transformer model is 65%, over 80% of its predictions are less than 1 unit away

TABLE 2. Table summarizing model performance by categorizing deviation of predictions from their true values where δ is the difference between the predicted and actual value.

Model	$\delta \leq 1$	$1 < \delta \leq 3$	$3 < \delta \leq 7$	$\delta > 7$
Dec.-only Transformer	79.19	11.72	7.81	1.26
Enc.-Dec. Transformer	81.14	10.72	7.40	0.72
Enc.-Dec. Transformer with Focal Loss	74.49	13.31	11.0	1.19
Enc.-Dec. Transformer with LSTM	80.34	11.03	7.66	0.96
Enc.-Dec. Transformer with CONV	80.24	11.14	7.70	0.90
LSTM	74.94	12.94	9.85	2.24
KNN	79.47	12.71	6.93	0.87
Rule-Based	54.95	36.97	7.64	0.42

from the true value. This is a strong result in the context of network performance, as scores like 9 and 10 often translate to a similar user experience.

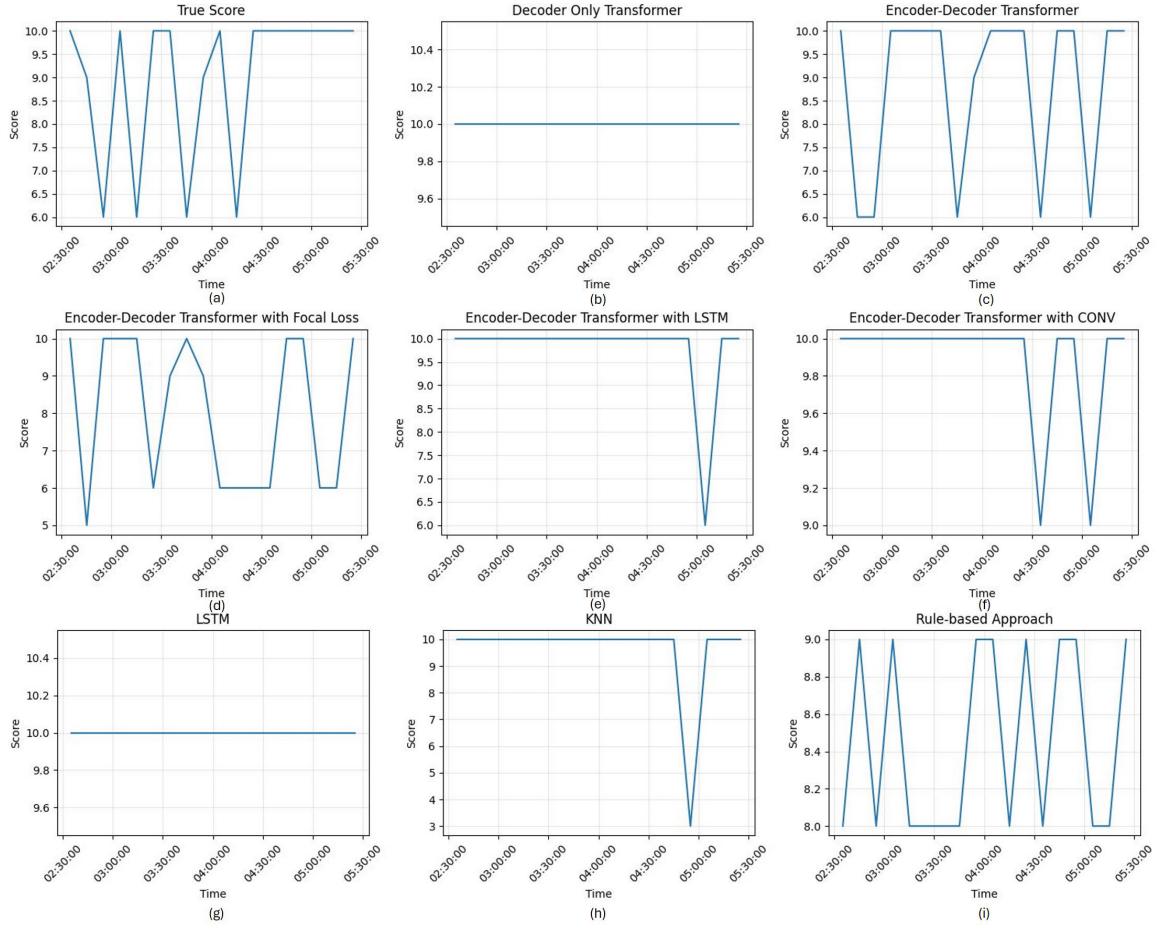


FIGURE 8. Plots visualizing the predictions of the model along sampled flight paths by categorizing the deviation of their predictions from the true values. The color of each point represents the magnitude of error. Green: $\delta \leq 1$, Yellow: $1 > \delta \leq 3$, Orange: $3 > \delta \leq 7$, Red: $\delta > 7$.

Additionally, we plot the predictions for 20 random flights on a map to provide a visual representation of the model's performance (Figure 7). The color of each sample represents the deviation of the prediction from the true value, corresponding to the ranges in Table 2. Green indicates a difference of less than or equal to 1, yellow represents differences between 1 and 3, orange corresponds to differences between 3 and 7, and red signifies differences greater than 7. The plots visually demonstrate that the proposed transformer model achieves the best performance, with a high concentration of green points and a minimal number of red ones. Specifically, 80% of its predictions are within 1 point of the true value, with only 0.86% exceeding a difference of 7 points.

We measured the inference time of the testing dataset for all the models, as shown in Table 3. The results show that the KNN model has a significantly higher inference time at 6353 seconds compared to the transformer-based and LSTM models, which maintain an inference time of 40 to 80 seconds. This high inference time is a fundamental limitation of KNN's instance-based learning approach, which requires a full pass through the entire dataset to find the

nearest neighbors for every single prediction. This makes the model's inference time directly dependent on the size of the training set. Therefore, while our study used only a subset of data, we conclude that the KNN model would be completely unscalable for global coverage, where the dataset would increase exponentially. In contrast, the transformer and LSTM models are highly efficient for inference, as their computational work is performed during training, allowing them to provide a fast and practical solution for real-world deployment. While the LSTM model has a faster inference time at 26.7s, our proposed transformer model is more accurate at 65% (compared to the LSTM's 61%). We consider the transformer's longer inference time of 42.05s a favorable trade-off for its superior accuracy.

Finally, to qualitatively evaluate our approach, we plot the time-series performance scores for a single flight in Figure 8, comparing the predictions of our proposed transformer model and the baselines against the true score. As shown in Figure 8 a, the ground truth performance begins at a score of 10, experiences several fluctuations down to 6, and concludes by returning to 10. The predictions from our proposed

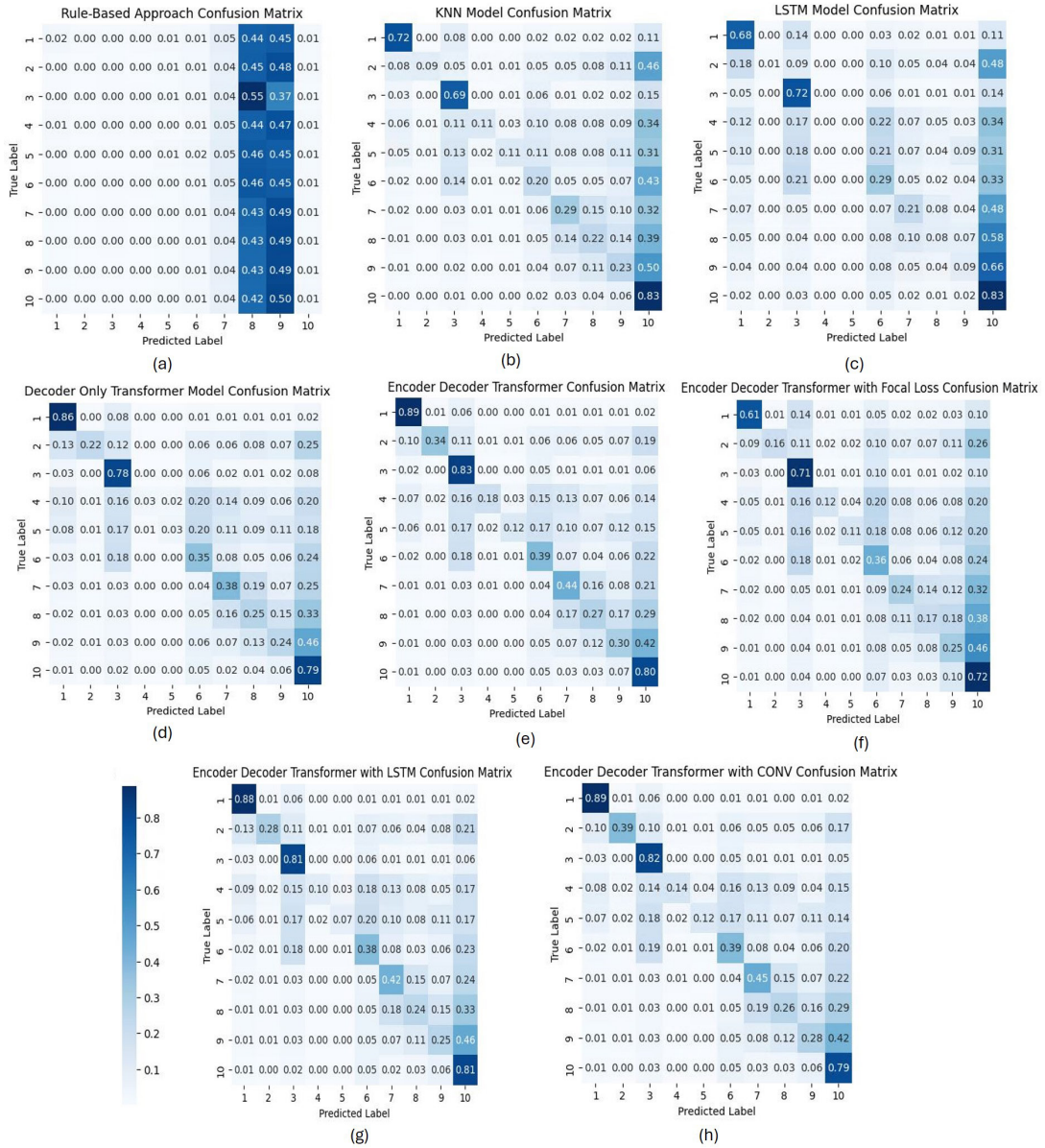


FIGURE 9. Confusion matrices for the proposed and benchmark models, providing a detailed, class-wise view of their predictive performance. Each matrix is labeled to show the results of the (a) Rule-Based approach, (b) KNN Model, (c) LSTM Model, (d) Decoder-Only Transformer, (e) Encoder-Decoder Transformer, (f) Encoder-Decoder Transformer with Focal Loss (g) Encoder-Decoder Transformer with LSTM, and (h) Encoder-Decoder Transformer with Convolution Layer.

transformer model in Figure 8c align well with this pattern, mirroring the general trend of performance degradation and recovery. However, there is a slight imprecision in the exact timing of these predicted events.

The baseline models demonstrate significant shortcomings in this scenario. Both the decoder-only transformer and the LSTM models, in Figure 8 b and g, fail to detect any performance issues, yielding a flat prediction of 10. Others, like the encoder-decoder with LSTM and the encoder-decoder with convolution, in Figure 8 e and f, predict only marginal or transient dips. The KNN model's output in Figure 8h is has

a sharp plunge to a score of 3. Whereas the encoder-decoder with focal loss model in Figure 8d tracks the overall trend but its predictions prolong the period of lower-rated performance unnecessarily.

Overall, this study successfully demonstrates a robust, transformer-based framework for predicting network performance in satellite-based avionic communication systems. We explored various transformer architectures and found that the encoder-decoder transformer offered the best balance between performance and complexity. While other variants, such as the decoder-only model, were less complex,

TABLE 3. Table summarizing model inference time.

Model	Inference Time
Dec.-only Transformer	37.7s
Enc.-Dec. Transformer	42.05s
Enc.-Dec. Transformer with Focal Loss	42.05s
Enc.-Dec. Transformer with LSTM	44.24s
Enc.-Dec. Transformer with CONV	78.61s
LSTM	26.7s
KNN	6353s

they did not achieve the same level of accuracy. Conversely, the more complex hybrid models with LSTM or CNN layers did not provide a notable performance improvement to justify their added complexity. Therefore, the encoder-decoder transformer stands out as the top performer across all metrics, proving its effectiveness in this domain.

The proposed transformer model is also preferred over the KNN model, which, despite its low RMSE, is not a practical solution. The transformer-based model excels at predicting the less-represented classes, which are critical for anomaly detection. It also has a lower false positive rate for the most frequent class (Class 10), providing more reliable predictions. Furthermore, its significantly lower inference time makes it a scalable solution for large-scale datasets. Its non-autoregressive nature is a key advantage, as it predicts the entire sequence at once, reducing latency and making it a practical solution for pre-flight planning. Our comprehensive evaluation, which considers both classification accuracy and prediction quality, provides a holistic overview of the model's performance, allowing for meaningful comparisons across different methodologies. While there is room for further improvement, particularly in addressing the challenges posed by underrepresented classes, the architecture shows significant promise and can be further enhanced with future fine-tuning methods.

VI. CONCLUSION

In this paper, we propose the use of transformer-based architecture for network performance prediction in satellite-based avionic communication systems. This is one of the first studies to utilize real-world avionic data for satellite-based avionic communication system optimization. We explore different transformer architectures, including the decoder-only transformer, the encoder-decoder transformer, and hybrid models combining encoder-decoder transformers with LSTM and Convolutional layers, to assess the potential benefits of additional feature extraction methods. Additionally, we compare our proposed approach against the performance of LSTM and KNN-based models through a comprehensive performance evaluation. The evaluation demonstrates that transformer models, particularly the encoder-decoder architecture, significantly outperform traditional approaches such as KNN and LSTM, both in classification accuracy and prediction quality.

Among the transformer variants, the encoder-decoder transformer stands out as the top performer, effectively capturing complex dependencies within the data and achieving the best sequence-level alignment. Despite challenges with underrepresented classes, the encoder-decoder transformer showcases substantial improvements over the KNN and LSTM models, making it a compelling solution for network performance prediction. The non-autoregressive nature of the encoder-decoder transformer further enhances its suitability, providing low-latency predictions while preserving the advantages of the attention mechanism. Our findings highlight the promise of transformers for network prediction tasks and suggest that further exploration of transformer architectures could lead to even greater advancements in performance, especially in handling data complexities and class imbalances.

While the weighted cross-entropy loss function was able to address the class imbalance for some classes and achieve some improvements, for future work we plan to explore additional fine tuning mechanisms to further improve the performance of the encoder-decoder transformer.

Our study focuses on U.S.-based flights utilizing GEO satellite networks to provide a proof of concept for our predictive framework. However, the proposed solution is highly transferable and can be extended to provide global coverage. The model's core architecture and methodology are not inherently tied to a specific geographic region or satellite type. The key is to train the model on a global dataset, which would be a logical next step in expanding the solution.

The framework can also be adapted to LEO/MEO constellations, which exhibit different dynamics due to more frequent satellite handovers. Adapting our solution to these constellations would require changes to our preprocessing pipeline, such as adjusting the handover probability algorithm, to accommodate their distinct network behaviors. This is a promising avenue for future work.

Additionally, our current model is designed for pre-flight network performance prediction. It provides a full flight path analysis at once, which is a non-autoregressive approach. However, this concept can be expanded to near real-time inference. Our solution could be modified to use an autoregressive transformer that predicts the network performance for the next time window, taking previous timestamps as input. This would allow for dynamic, in-flight re-evaluation and adjustment of predictions, bridging the gap between proactive planning and reactive optimization. The emphasis on low inference time and scalability, as demonstrated by the performance of our transformer models, is a crucial step towards making this a practical solution for the industry. Our approach provides a flexible tool that can be adapted for a variety of operational scenarios and network types.

REFERENCES

- [1] *Whitepaper: Future Aircraft Communications*, Int. Air Transp. Assoc. (IATA), Montreal, QC, Canada, Jun. 2024.

- [2] A. P. T. Adithyababu, F. Boulos, and S. Caizzzone, "Avionic satellite communication terminal requirements and technology maturity analysis," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 39, no. 1, pp. 4–17, Jan. 2024.
- [3] SatcomDirect. (Mar. 2023). *Aero Connectivity Considerations in a Multi-Orbit Landscape*. [Online]. Available: <https://www.satcomdirect.com/wp-content>
- [4] H. Mukhtar, R. Schaub, and M. Erol-Kantarci, "SkyNetPredictor: Network performance prediction in avionic communication using AI," 2025, *arXiv:2504.14443*.
- [5] F. Fourati and M.-S. Alouini, "Artificial intelligence for satellite communication: A review," *Intell. Converged Netw.*, vol. 2, no. 3, pp. 213–243, Sep. 2021.
- [6] A. Haldorai, R. B. Lincy, M. Suriya, and M. Balakrishnan, "Satellite-terrestrial integrated computing and artificial intelligence as a means of achieving handover management," in *Proc. IEEE Int. Conf. Comput., Power Commun. Technol. (IC2PCT)*, Feb. 2024, pp. 874–877.
- [7] P. Henarejos, M. À. Vázquez, and A. I. Pérez-Neira, "Deep learning for experimental hybrid terrestrial and satellite interference management," in *Proc. IEEE 20th Int. Workshop Signal Process. Adv. Wireless Commun. (SPAWC)*, Jul. 2019, pp. 1–5.
- [8] F. G. Ortiz-Gomez et al., "Machine learning for radio resource management in multibeam GEO satellite systems," *Electronics*, vol. 11, no. 7, p. 992, Mar. 2022. [Online]. Available: <https://www.mdpi.com/2079-9292/11/7/992>
- [9] M. O. K. Mendonca et al., "Flexible payload configuration for satellites using machine learning," 2023, *arXiv:2310.11966*.
- [10] V.-P. Bui, T. Van Chien, E. Lagunas, J. Grotz, S. Chatzinotas, and B. Ottersten, "Learning to optimize: Balancing two conflict metrics in MB-HTS networks," 2022, *arXiv:2208.13205*.
- [11] V. Rizzello, B. Böck, M. Joham, and W. Utschick, "Reverse ordering techniques for attention-based channel prediction," 2023, *arXiv:2302.00341*.
- [12] H. Mukhtar, R. Schaub, and M. Erol-Kantarci, "Air traffic and usage predictions in avionic communications using attention based VAEGAN model," in *Proc. IEEE Int. Conf. Mach. Learn. Commun. Netw. (ICMLCN)*, May 2024, pp. 220–226.
- [13] G. Greco et al., "Satellite communications for aviation: Challenges and future directions," *IEEE Access*, vol. 8, pp. 195531–195550, 2020. [Online]. Available: <https://ieeexplore.ieee.org/document/9190152>
- [14] J. Gonzalez et al., "Satellite communication for aviation: A comprehensive overview of GEO, MEO, and LEO systems," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 3, pp. 1670–1682, Mar. 2021. [Online]. Available: <https://ieeexplore.ieee.org/document/9392999>
- [15] A. Gannon, B. Schoenholz, J. Downey, and J. Clapham, "Continuous spacecraft communications via make-before-break antenna array beam steering," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Jun. 2024, pp. 1837–1842.
- [16] R. Prasad and M. Ruggieri, *Applied Satellite Navigation Using GPS, GALILEO, and Augmentation Systems*. Norwood, MA, USA: Artech House, 2003.
- [17] Intelsat. (2024). *Satellite Connectivity Solutions for Commercial Airlines*. Accessed: Feb. 4, 2025. [Online]. Available: <https://pages.intelsat.com/rs/409-OAJ-662/images/BR-Satellite-Connectivity-Solutions.pdf>
- [18] C. E. Shannon, "A mathematical theory of communication," *Bell Syst. Tech. J.*, vol. 27, no. 3, pp. 379–423, Jul. 1948. [Online]. Available: <https://ieeexplore.ieee.org/document/6773024>
- [19] Army Technology. (2021). *Resilience and Reliability Where It's Needed Most: Flexmove for Government*. Accessed: Mar. 2, 2025. [Online]. Available: <https://www.army-technology.com/sponsored/resilience-and-reliability-where-its-needed-most-flexmove-for-government/>
- [20] A. Goldsmith, *Wireless Communications*. Cambridge, U.K.: Cambridge Univ. Press, 2005.
- [21] A. Vaswani et al., "Attention is all you need," 2017, *arXiv:1706.03762*.
- [22] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," 2016, *arXiv:1607.06450*.
- [23] M. Bhattacharyya. (May 2020). *Gradient Accumulation: Overcoming Memory Constraints in Deep Learning*. [Online]. Available: <https://towardsdatascience.com/gradient-accumulation-overcoming-memory-constraints-in-deep-learning>
- [24] A. Paszke et al., "Automatic differentiation in Pytorch," in *Proc. NIPS*, 2017, pp. 1–4.
- [25] T. Dao, D. Y. Fu, S. Ermon, A. Rudra, and C. Ré, "FlashAttention: Fast and memory-efficient exact attention with IO-awareness," 2022, *arXiv:2205.14135*.
- [26] K. Pearson, "Note on regression and inheritance in the case of two parents," *Proc. Roy. Soc. London*, vol. 58, nos. 347–352, pp. 240–242, Dec. 1895.



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